Problems with Chinchilla Approach 2: Systematic Biases in IsoFLOP Parabola Fits

Eric Czech
Open Athena AI Foundation

Abstract

Chinchilla Approach 2 is arguably the most widely adopted method for fitting neural scaling laws in practice. The parabolic approximation inherent to this method introduces systematic biases in compute-optimal allocation estimates, even on noise-free synthetic data under ideal experimental conditions. Three sources of error are examined: IsoFLOP sampling grid width, which affects Taylor approximation accuracy of parabolic fits; uncentered IsoFLOP sampling, which distorts scaling intercepts or exponents depending on how offsets vary with compute budget; and loss surface asymmetry ($\alpha \neq \beta$), which attenuates or amplifies other biases. Effects of these errors are demonstrated as misallocations at open-model frontier compute scales. Namely, Approach 2 extrapolations from published Llama 3 IsoFLOP data suggest an underallocation of parameters and that 6–10% (\$1.3–2.2M USD) of the 3.8×10^{25} FLOPs training budget could have been saved to reach the same loss. Simulated misallocations for multimodal models are also presented and show an even greater potential for this kind of opportunity cost due to their more asymmetric loss surfaces. Chinchilla Approach 3 largely eliminates this potential, but is often regarded as less data-efficient, unstable, prone to local minima, and typically more complex to implement. Each of these concerns are shown to be unfounded or addressable, especially when the partially linear structure of the Approach 3 optimization objective is exploited through Variable Projection. This enables unbiased, parametric inference on all five loss surface parameters as a two-dimensional optimization that is well-conditioned, analytically differentiable, and highly amenable to dense, or even exhaustive, grid search. It may serve as a more convenient replacement for Approach 2 or a more scalable alternative for adaptations of Approach 3 to richer scaling law formulations.

1 Introduction

The Chinchilla paper [Hoffmann et al., 2022] introduced three approaches to scaling law estimation, and of these, Approach 2 appears to have seen the broadest adoption. It has been used by leading AI labs including DeepMind [Hoffmann et al., 2022, Alabdulmohsin et al., 2023] (its creators), Meta [Grattafiori et al., 2024, Tay et al., 2025], DeepSeek [Bi et al., 2024], Microsoft [Jiang et al., 2025], Amazon [Haldar and Pinto, 2023], Waymo [Chen et al., 2025], and Arc Institute [Nguyen et al., 2024], among others. It is also a workhorse method for academic scaling law studies [Li et al., 2024, Nie et al., 2025, Wagh et al., 2024] and tutorials[1] like those from Andrej Karpathy.

The method's appeal lies in its putative stability and data efficiency relative to nonlinear optimization over all loss surface parameters. Rather than fitting all five parameters of the loss surface simultaneously, Approach 2 targets only the two scaling exponents, relying on second-order Taylor approximations that reduce each IsoFLOP curve to a simple parabola. This sacrifices recovery of the full loss surface but makes estimation simple and stable, letting practitioners extract the

[1] https://github.com/karpathy/nanochat/discussions/420

most actionable quantities for compute allocation planning through a sequence of straightforward polynomial and linear fits, without ever needing a nonlinear optimizer.

Despite this broad adoption, the sensitivity of the method's core approximations and its behavior on loss surfaces that are less symmetric than the original Chinchilla form (where parameter and token scaling exponents are roughly equal) have not, to our knowledge, been studied in detail. Here we revisit the basics of how to apply a simple model like Chinchilla with high precision and stability. We investigate this through noise-free synthetic simulations that isolate systematic biases inherent to Approach 2 by eliminating all sources of statistical noise.

We show how these biases affect allocation planning for final training runs at large compute budgets. We show how extrapolation errors trace back to suboptimal IsoFLOP experiment design, and that pathologies in these designs can be observed in real, high-profile scaling law studies even if they are difficult to quantify precisely. Finally, we propose an alternative fitting method that is simple, stable, and free of these biases while building on the same intuitive computational shortcut: optimizing exponential terms separately from linear terms. We call this approach Variable Projection with Non-negative Least Squares (VPNLS).

This investigation is also motivated by a broader landscape of *analytical* extensions to the Chinchilla loss surface. A growing body of work adds or modifies terms in the original functional form to account for additional training configuration choices such as data repetition [Muennighoff et al., 2023], overfitting [Yang et al., 2024], precision [Kumar et al., 2024], optimizers [Volkova et al., 2026], MoE sparsity [He et al., 2025], pruning [Jin et al., 2025], data quality [Subramanyam et al., 2025], data mixtures [Shukor et al., 2025a, Bandarkar et al., 2025], model shape/context length [Cheng and Lucchi, 2025], non-embedding parameters [Sardana and Frankle, 2024], and downstream task performance [Bhagia et al., 2024], to name a few. These extensions prescribe explicit functional forms rather than inferring scaling law structure automatically, and they build directly on the Chinchilla model as a foundation. Similar studies [Bi and Calhoun, 2025, Goyal et al., 2024, Tian et al., 2025, Hashimoto et al., 2021] extend individual terms in isolation (e.g. token scaling terms alone) or propose modified Kaplan scaling laws [Kaplan et al., 2020], both of which would be relevant for comparable Chinchilla formulations as well. A fitting method that recovers the base surface with greater scalability and stability may therefore offer a stronger starting point for these adaptations.

2 Preliminaries

Neural scaling laws describe how model performance improves with compute. The Chinchilla loss surface models this relationship as:

$$L(N, D) = E + \frac{A}{N^\alpha} + \frac{B}{D^\beta} \tag{1}$$

where N is the number of parameters, D is the number of training tokens, E is the irreducible loss, and A, B, α, β capture how quickly performance improves with scale.

Given a compute budget $C \approx 6ND$, the optimal allocation satisfies:

$$N^* \propto C^a \quad \text{where} \quad a = \frac{\beta}{\alpha + \beta} \tag{2}$$

$$D^* \propto C^b \quad \text{where} \quad b = \frac{\alpha}{\alpha + \beta} \tag{3}$$

Recovering just the exponents a and b from empirical training runs is useful for a variety of purposes. Two canonical approaches for this exist:

2.1 Approach 2: IsoFLOP Parabolic Fitting

This approach takes advantage of the fact that along a fixed-compute contour (IsoFLOP curve), loss as a function of $\log N$ is approximately parabolic near the optimum. The procedure has three steps:

1. **Sample IsoFLOP contours:** For each compute budget C, train models at various (N, D) pairs satisfying $C = 6ND$.
2. **Fit parabolas:** For each budget, fit $L = p(\log N)^2 + q(\log N) + r$ and extract the minimum N^*.
3. **Fit power laws:** Regress $\log N^*$ against $\log C$ to recover the exponent a (and similarly for D^*, b).

The appeal is simplicity: only polynomial fits, no nonlinear optimization. The parabolic approximation comes from a Taylor expansion of the loss surface around the optimum.

2.2 Approach 3: Direct Surface Fitting

The alternative is to fit all five parameters (E, A, B, α, β) simultaneously by minimizing the residual sum of squares (RSS) directly:

$$\min_{E,A,B,\alpha,\beta} \sum_i \big(L_i - \hat{L}(N_i, D_i)\big)^2 \quad \text{where} \quad \hat{L}(N, D) = E + \frac{A}{N^\alpha} + \frac{B}{D^\beta} \tag{4}$$

At least two practical issues arise with this direct formulation. First, E, A, and B must remain positive for the surface to be physically meaningful, requiring box constraints. Second, the prediction involves a sum of exponentials that can span many orders of magnitude, creating numerical optimization difficulties.

The Chinchilla paper [Hoffmann et al., 2022] addresses both issues with an adapted formulation. A LogSumExp (LSE) reparameterization optimizes (e, a, b, α, β) where $E = e^e$, $A = e^a$, $B = e^b$, enforcing positivity without explicit bounds. The predicted log-loss is computed as:

$$\log \hat{L}(N, D) = \text{logsumexp}\big(e,\ a - \alpha \log N,\ b - \beta \log D\big) \tag{5}$$

The objective is then a Huber loss on log-predictions rather than MSE on predictions. Operating in log-space improves numerical conditioning, and the Huber function provides robustness to outliers from noisy training runs.

Because we evaluate methods on simulated data with known noise models and want maximum likelihood estimates in some configurations, we use MSE rather than Huber throughout this study. The LSE reparameterization and log-space objective are evaluated directly in the exponent inference comparison (Section 7.4).

3 Symmetric Surfaces: Unbiased Estimation in Ideal Conditions

Before examining failure modes, we establish that Approach 2 works perfectly under ideal conditions. Consider a **symmetric** loss surface where $\alpha = \beta$:

$$L(N, D) = 1.69 + \frac{400}{N^{0.31}} + \frac{400}{D^{0.31}} \tag{6}$$

With equal exponents, the optimal allocation splits compute evenly between parameters and data. The true scaling exponents are:

$$a = b = \frac{0.31}{0.31 + 0.31} = 0.5 \tag{7}$$

We sample five IsoFLOP contours spanning 10^{17} to 10^{21} FLOPs, with 15 model sizes per curve, fit parabolas to each, and extract the optimal token count D^*. Most simulations in this study use these same five compute budgets and 15 points per IsoFLOP curve, unless stated otherwise.

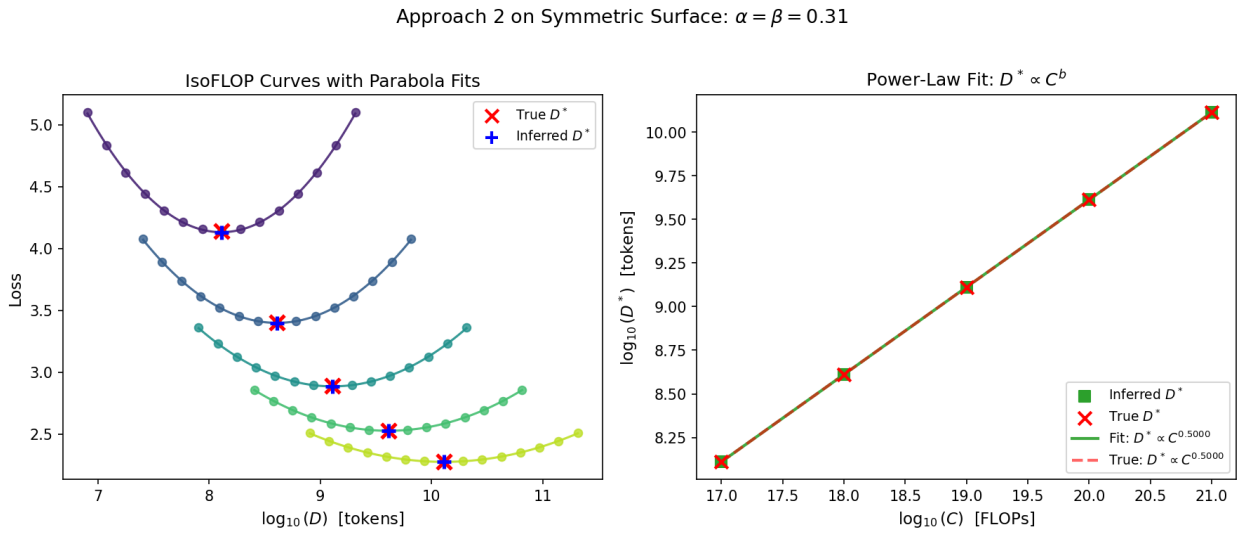


Figure 1: Approach 2 applied to a symmetric loss surface. Left: IsoFLOP curves with fitted parabolas. True (×) and inferred (+) optima are indistinguishable. Right: Power-law fit recovers the exact scaling exponent.

The results confirm perfect recovery of the token scaling exponent and intercept as shown in Figure 1 and Table 1. This establishes our baseline. Approach 2 is precisely correct under ideal conditions that are unrealistic in practice. The problems arise when we deviate from these ideal conditions, as we show in the following sections where these conditions are perturbed in controlled ways.

Parameter	True Value	Inferred Value	Relative Error
b (D^* exponent)	0.500000	0.500000	$+6.2 \times 10^{-12}\%$
b_0 (D^* intercept)	-0.389076	-0.389076	$-1.4 \times 10^{-10}\%$

Table 1: Approach 2 parameter recovery on the symmetric surface.

Summary. On a symmetric loss surface with perfectly crafted IsoFLOP grid sampling, Approach 2 recovers both exponents and intercepts with machine-precision accuracy. When $\alpha = \beta$, the parabola vertex shift is zero, so the inferred optima coincide with the true optima.

4 Asymmetric Surfaces: Intercept and Extrapolation Errors

We repeat the exact same procedure as before: perfect sampling centers, no noise, identical methodology. The only change is that the loss surface is now **asymmetric** ($\alpha \neq \beta$).

Simulation results show that when the loss surface is asymmetric, Approach 2 produces systematically wrong intercepts while exponents remain accurate. This is not statistical noise; it is a deterministic bias from fitting parabolas to a non-parabolic surface.

We test two configurations to see how the effect scales:

- **Chinchilla:** $\alpha = 0.34$, $\beta = 0.28$ (ratio ≈ 1.2)
- **Asymmetric:** $\alpha = 0.46$, $\beta = 0.15$ (ratio $= 3.0$)

This Asymmetric surface definition is not unrealistic. An exponent ratio of 3.0 is comparable to what has been observed in practice. Kaplan et al. [Kaplan et al., 2020] reported compute-optimal allocation exponents of $a = 0.73$, $b = 0.27$ (as cited in DeepSeek [Bi et al., 2024]). This implies a loss surface exponent ratio of $\beta/\alpha \approx 2.7$. The asymmetry runs in the opposite direction from our Asymmetric surface ($\beta > \alpha$ rather than $\alpha > \beta$), but the degree of imbalance is similar, and it is the magnitude of the imbalance, not its direction, that drives the biases studied here.

Parameter	True Value	Inferred Value	Relative Error
b (D^* exponent)	0.548387	0.548387	$\approx 0\%$
b_0 (D^* intercept)	-0.555357	-0.578092	-4.1%

Table 2: Approach 2 parameter recovery on the Chinchilla surface.

Parameter	True Value	Inferred Value	Relative Error
b (D^* exponent)	0.750000	0.750000	$\approx 0\%$
b_0 (D^* intercept)	-1.345791	-1.459957	-8.5%

Table 3: Approach 2 parameter recovery on the Asymmetric surface.

Figure 2 shows the results from this simulation, with relative errors in Tables 2 and 3. A few percent error in the intercept might seem minor, but consider that this simulation provided every advantage other than loss surface asymmetry. The data contains no measurement noise and the sampling is perfect, with IsoFLOP grids centered precisely at the true optimum (something difficult or impossible to know in practice). And the loss surface parameters are standard in the Chinchilla case, taken directly from the Chinchilla paper.

Summary. Even under unrealistically ideal conditions, Approach 2 produces biased intercepts for asymmetric surfaces. The error is systematic, a property of the parabolic approximation, not statistical noise.

4.1 Underlying Causes

The IsoFLOP loss curve is not a true parabola. When a parabola is fit to this curve, the parabola's minimum (vertex) does not land exactly at the true optimum. It shifts slightly, and this shift

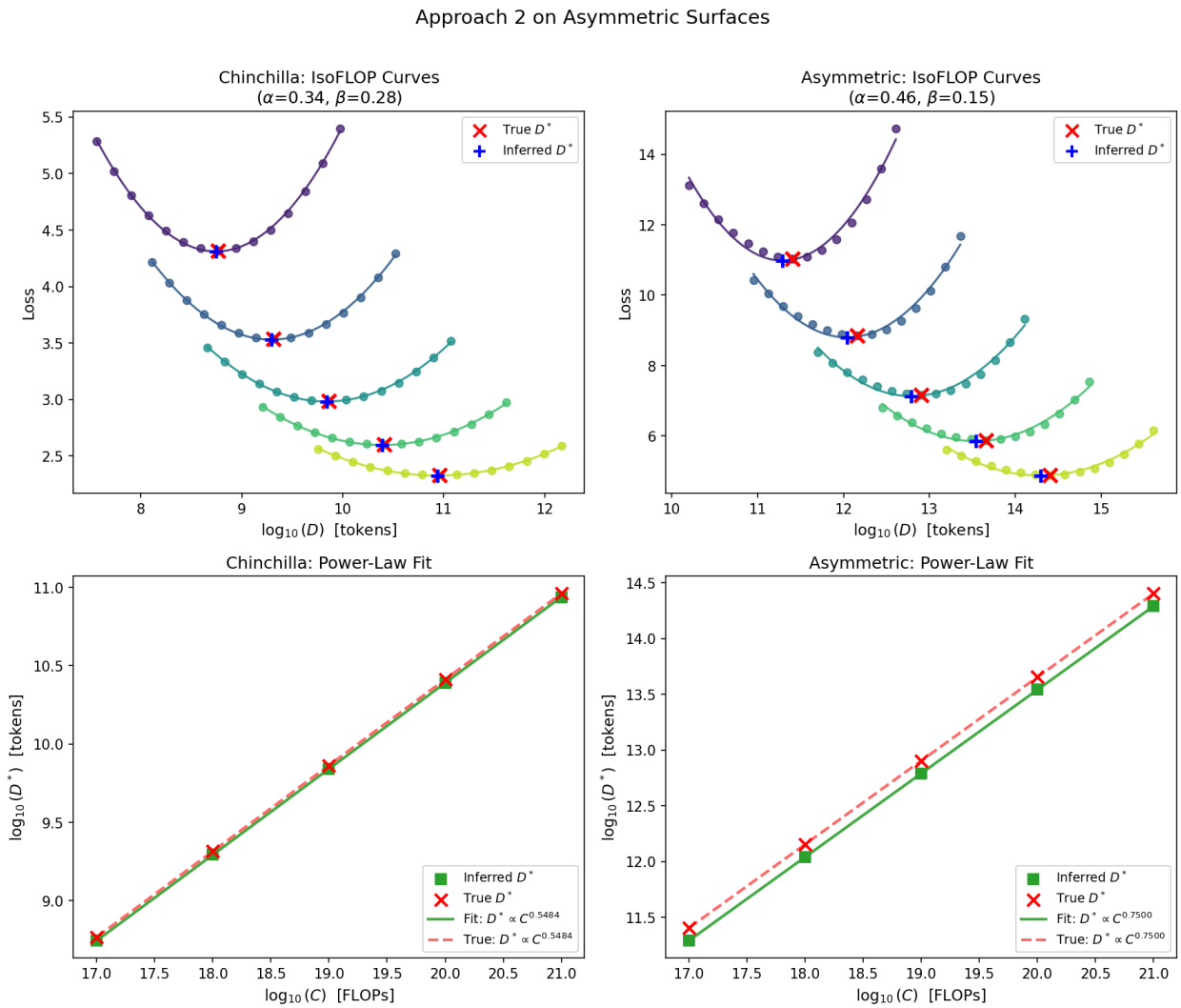


Figure 2: Approach 2 on asymmetric loss surfaces. Note the visible gap between true (dashed) and inferred (solid) power-law lines in the Asymmetric case. The exponents match perfectly, but the intercepts differ.

depends only on the loss surface shape (α, β) and the sampling grid. It does not depend on compute budget. The sampling grid size becomes important here as wider grids amplify the mismatch between the true curve and its parabolic approximation, increasing the vertex shift.

Because the IsoFLOP parabola is fit in $\log N$ space (as described in the Approach 2 procedure), the vertex shift directly biases N^*. Since $C = 6ND$, analyzing the bias in either N^* or D^* is sufficient; we focus on N^* here since that is where the parabolic fit typically operates.

Since the vertex shift is constant across all compute budgets, it biases every inferred N^* by the same multiplicative factor. When fitting $\log N^*$ vs $\log C$ to extract scaling exponents:

- The **slope (exponent)** is unchanged: multiplying all N^* values by a constant factor adds a constant to $\log N^*$, which does not affect the slope.
- The **intercept** absorbs the entire error, biased by exactly that multiplicative factor.

The intercept error can be derived analytically in closed form. The parabola vertex shifts by

δw (in log-space), giving an intercept error of:

$$\text{Intercept error} = 10^{\delta w} - 1 \tag{8}$$

where $\delta w = f(\alpha, \beta, W, n)$ depends only on the surface exponents and the sampling grid (width W in log-space, number of points n per IsoFLOP curve), not on C, E, A, or B. Here W spans $10^{-W/2}$ to $10^{W/2}$ times the optimal N^*, so $W = 2.41$ (the XL grid) means sampling from $\frac{1}{16}\times$ to $16\times$ the optimum. Key properties:

- $\delta w = 0$ when $\alpha = \beta$ (symmetric surfaces have no error)
- δw grows with $|\alpha - \beta|$ (more asymmetry, more error)
- δw grows with W (wider sampling range, more error)

For example, with the Chinchilla parameters ($\alpha = 0.34$, $\beta = 0.28$): the XS grid ($W = 0.60$) yields 0.3% intercept error, while the XL grid ($W = 2.41$) yields 4.1% error.

The full derivation[2] provides the closed-form expression for vertex shift δw as a function of α, β, W, and n. It also shows how this shift translates directly into intercept error, independent of compute budget.

Intuition via Taylor expansion. A parabola is a 2nd-order polynomial, equivalent to a 2nd-order Taylor expansion around the optimum. The approximation $L(w) \approx L(0) + \frac{1}{2}L''(0)w^2$ is only valid when higher-order terms are negligible, i.e., when samples are close to the true minimum. As sampling range increases, 3rd and 4th order terms grow. For symmetric surfaces ($\alpha = \beta$), odd-order terms cancel by symmetry, preserving the vertex location. For asymmetric surfaces, they do not cancel, shifting the fitted vertex away from the true optimum.

4.2 Error Implications

Extrapolation to higher compute budgets requires both exponents and intercepts to be correct. The previous subsection (Section 4.1) established that asymmetric loss surfaces produce provably biased intercepts even under ideal experimental conditions. Here we quantify what those errors mean in practical terms by examining compute-optimal token prediction: given a compute budget, how many tokens does the inferred scaling law predict?

Up to this point, all analysis has assumed a single fixed sampling grid width. We now examine how token prediction error varies with both compute budget and sampling grid width. For surfaces with asymmetric exponents, wider sampling grids amplify the parabola-fitting mismatch, increasing the constant vertex shift and thus the intercept bias. To make this comparison concrete, we first define what "wider" and "narrower" mean in quantitative terms.

A sampling grid of "$\pm k\times$" means the sampled values (whether model sizes or token counts) range from $\frac{1}{k}$ to k times the true optimum at each compute budget. The total range covered is k^2 (the ratio of largest to smallest), and the $\log_{10}$ of that ratio gives the number of decades the grid spans end-to-end. Table 4 shows the four grid widths used in this analysis.

The key observations from Figure 3 (raw data in Appendix A) are:

- **Symmetric surfaces are unaffected:** When $\alpha = \beta$, all grid widths produce zero error.
- **Asymmetric surfaces underestimate:** Negative errors mean the inferred D^* is smaller than the true D^*. Following these predictions would undertrain the model.

[2] https://github.com/Open-Athena/scaling-law-analysis/blob/main/results/article/static/scaling_parameter_errors.pdf

Grid Name	$\pm k\times$	Sampling Range	Total Ratio	Decade Span
Extra Small (XS)	$\pm 2\times$	$\frac{1}{2}\times$ to $2\times$	$4\times$	0.60
Small (S)	$\pm 4\times$	$\frac{1}{4}\times$ to $4\times$	$16\times$	1.20
Large (L)	$\pm 8\times$	$\frac{1}{8}\times$ to $8\times$	$64\times$	1.81
Extra Large (XL)	$\pm 16\times$	$\frac{1}{16}\times$ to $16\times$	$256\times$	2.41

Table 4: Sampling grid widths used throughout this paper. Real experiments typically span 1–2 decades, placing the Small and Large grids within the realistic range. The Extra Large grid ($\pm 16\times$, $\sim$2.4 decades) is the default used in all single-grid analyses in the preceding sections.

- **Wider grids amplify error:** Moving from XS ($\pm 2\times$) to XL ($\pm 16\times$) grids increases error from 0.3% to 5.1% on Chinchilla, and from 1.7% to 23% on the Asymmetric surface.
- **Asymmetry further amplifies error:** The Asymmetric surface ($\alpha/\beta = 3$) shows roughly 4–5$\times$ larger errors than Chinchilla at each grid width.

Summary. Consider the Chinchilla surface with the Large grid ($\pm 8\times$), a practical sampling range for real experiments. When extrapolating to 10^{24} FLOPs, the true optimal token count is 4.04 trillion, but Approach 2 predicts only 3.92 trillion: a 2.9% underestimate, or roughly 117 billion fewer tokens than optimal. While 2.9% may seem modest, recall that this simulation uses unrealistically ideal conditions: perfectly centered sampling grids at every compute budget and zero measurement noise. Real experiments, where the true optimum is unknown, data is noisy, and the scaling exponent imbalance may be larger than Chinchilla's modest $\alpha/\beta \approx 1.2$, can only do worse.

5 Off-Center Sampling: Exponent and Extrapolation Errors

The previous sections assumed perfectly centered sampling. At every compute budget, the IsoFLOP grid was placed exactly at the true optimum. In practice, N^* is not known before running the experiment. Sampling centers are guesses, informed by prior estimates or heuristics, and they will likely be wrong by some amount.

This is a distinct source of error from the asymmetry bias examined earlier. Asymmetry errors arise from the shape of the loss surface ($\alpha \neq \beta$); off-center errors arise from where the sampling grid is placed. To isolate this new effect, we return to the symmetric surface ($\alpha = \beta = 0.31$) where asymmetry bias is zero by construction.

5.1 Constant Multiplicative Bias

The simplest form of off-center sampling is a constant multiplicative offset: every compute budget's sampling center is shifted by the same factor from the true optimum. A "3$\times$ offset" means each IsoFLOP grid is centered at $3 \times D^*$ instead of D^*, so the grid midpoint consistently sits at three times the true optimal token count.

Because this offset is the same at every compute budget, it has a familiar geometric effect where each parabola vertex shifts by a constant amount in log-space. This is the same mechanism as asymmetry bias. The slope of $\log D^*$ vs $\log C$ is unaffected (a constant additive shift in log-space does not change the slope), so the scaling exponent is preserved perfectly. The intercept, however, absorbs the entire error.

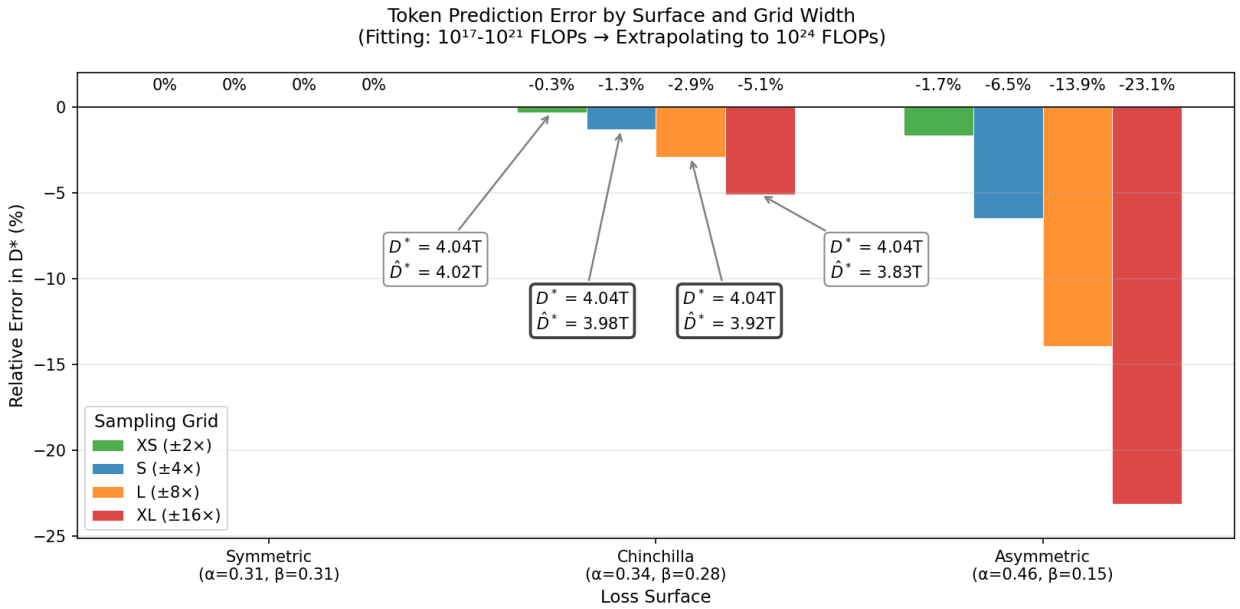


Figure 3: Relative error in compute-optimal token prediction when extrapolating from the training range (10^{17}–10^{21} FLOPs) to 10^{24} FLOPs. Negative values indicate underestimation: the inferred scaling law predicts fewer tokens than optimal. Bars are grouped by sampling grid width. Annotations for the Chinchilla surface show D^* (true compute-optimal token count) versus $\hat{D}^*$ (the Approach 2 estimate); the Small and Large grid annotations are emphasized as they fall within the realistic 1–2 decade range typical of scaling law experiments.

The extrapolation bar chart (top right of Figure 4) shows what this means for token prediction. All four grid widths overestimate D^*, with the narrowest grid (XS) producing the largest error. This is the reverse of the asymmetry bias pattern, where wider grids amplified error. Here, narrower grids are more sensitive to off-center placement because fewer samples lie near the true optimum.

The intercept error panel (bottom right) confirms the pattern across the full continuum of grid widths. The error is always positive (the inferred D^* overshoots) and decreases monotonically as the grid widens, reflecting how a wider sampling range brings more of the true loss curve's shape into the fit, partially compensating for the misplaced center.

Summary. Consider the symmetric surface with the Large grid ($\pm 8\times$) and a $3\times$ offset, where every IsoFLOP grid is centered at three times the true optimal token count. When extrapolating to 10^{24} FLOPs, the true optimal token count is 408.2 billion, but Approach 2 predicts 419.0 billion: a 2.6% overestimate, roughly 10.8 billion more tokens than optimal. Compare this with the Chinchilla asymmetry result at the same grid width: a 2.9% underestimate. The magnitudes are comparable, but the sources are entirely different. Asymmetry bias comes from the shape of the loss surface; off-center bias comes from where the grid is placed. In a real experiment, both act simultaneously.

5.2 Drifting Bias

When the offset varies with compute budget, a different failure mode occurs, and we illustrate this by applying a multiplicative drift. The sampling center starts at the true optimum for the lowest budget and drifts to $3\times$ the true optimum at the highest budget, interpolating linearly in log-compute space.

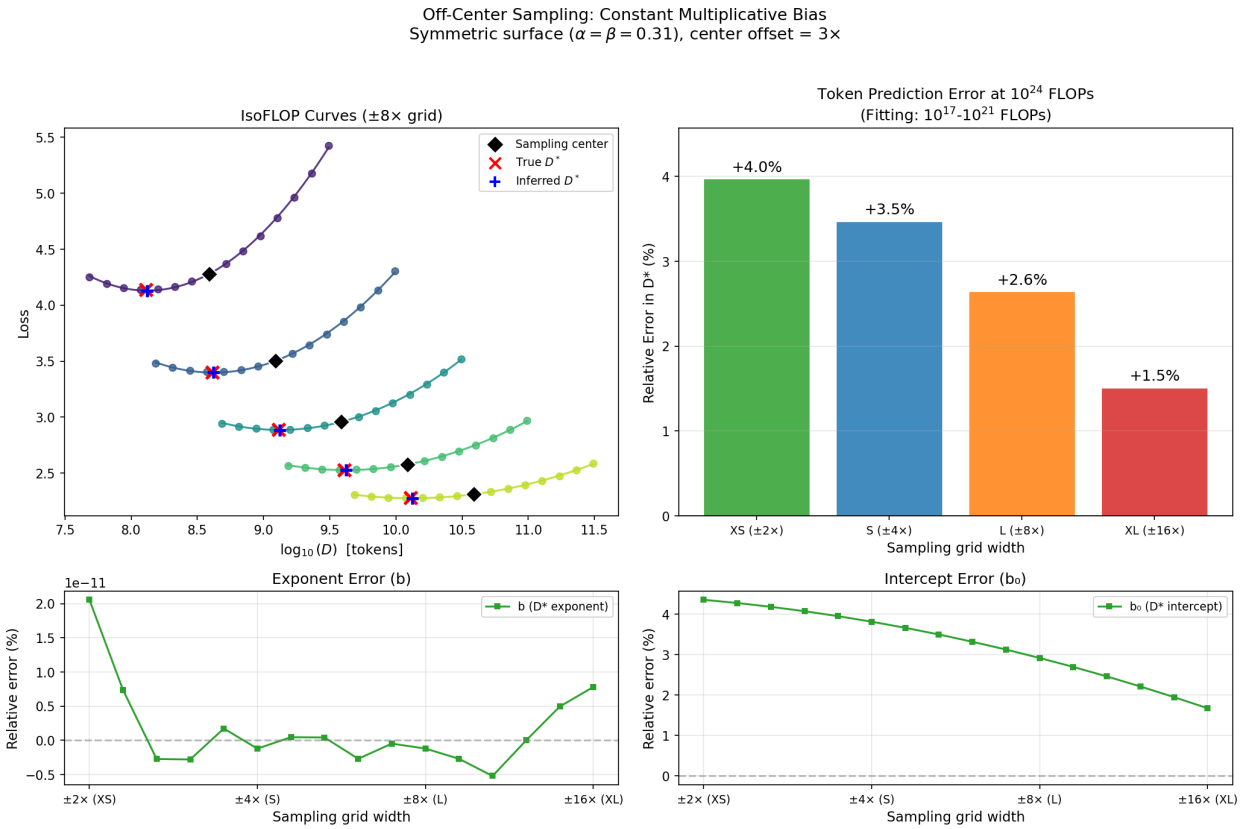


Figure 4: Effect of a constant 3× offset in sampling centers on the symmetric surface. Top left: IsoFLOP curves at the Large grid (±8×), with black diamonds marking the (off-center) sampling center, red × the true D^*, and blue + the inferred D^*. Top right: extrapolation error in compute-optimal token prediction at 10^{24} FLOPs for each grid width. Bottom row: exponent and intercept errors across grid widths from XS (±2×) to XL (±16×), plotted on the same y-axis scale. The exponent is recovered perfectly (flat at zero) while the intercept shows systematic bias that varies with grid width.

Because the offset now differs across compute budgets, it no longer cancels in the slope of $\log D^*$ vs $\log C$. Both the exponent and the intercept are affected.

The bottom-left panels of Figures 4 and 5 show the most important difference between the two. Those plots demonstrate how constant offsets in sampling centers do not bias exponent inference while multiplicative offsets do.

Summary. Constant bias preserves exponents and any compute-dependent bias pattern distorts them. The distinction matters because exponent errors compound during extrapolation, while intercept errors remain fixed.

6 Real IsoFLOP Curves: Evidence from Published Studies

The previous sections used synthetic, noise-free simulations to isolate Approach 2's biases under controlled conditions. A natural question is whether the conditions that trigger these biases, asymmetric loss surfaces and imperfectly centered sampling, actually arise in practice. To get a sense of this, we can look at IsoFLOP curves published in three of the most prominent scaling law studies [Hoffmann et al., 2022, Grattafiori et al., 2024, Bi et al., 2024] in Figure 6; additional examples

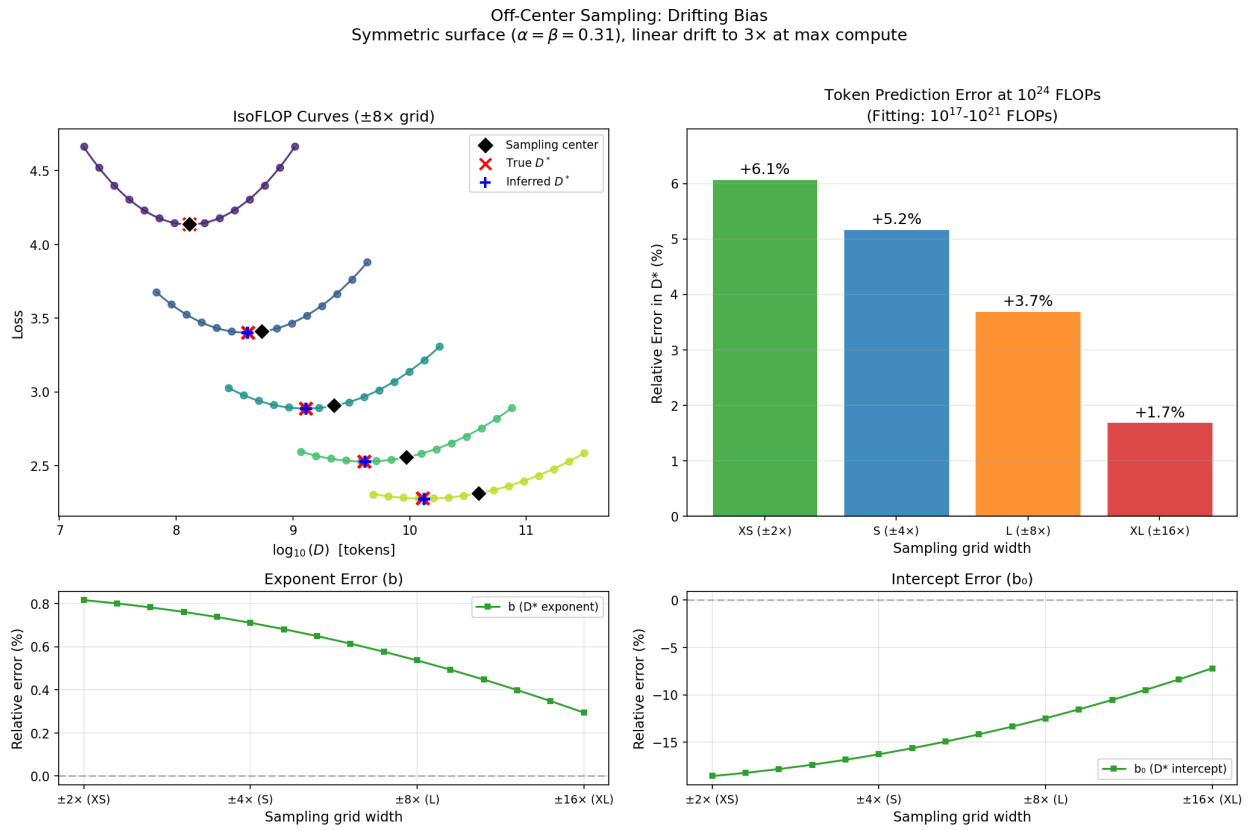


Figure 5: Effect of a linear drift in sampling centers (centered at true optimum for lowest budget, drifting to 3× at highest budget) on the symmetric surface. Unlike the constant bias case, the exponent error (bottom left) is now non-zero: the slope of $\log D^*$ vs $\log C$ is distorted because the offset varies across compute budgets.

are collected separately[3].

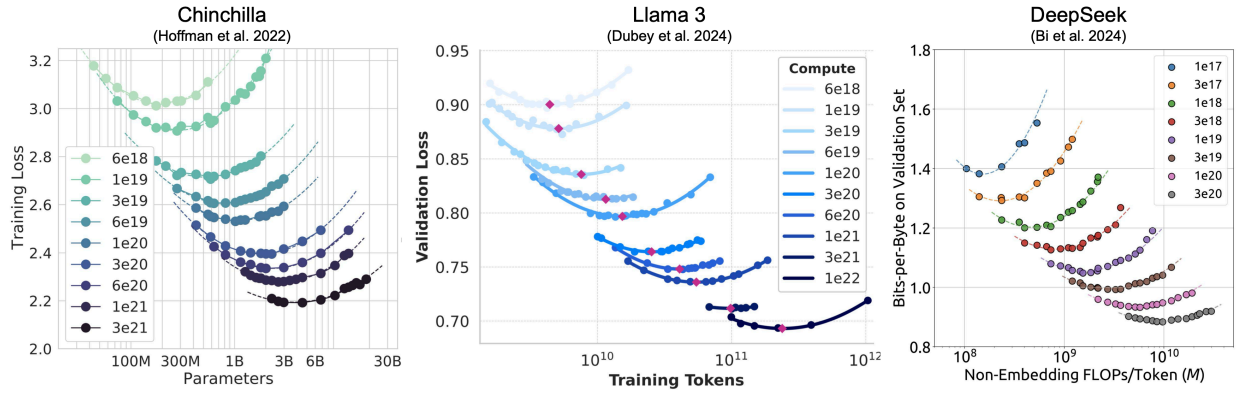


Figure 6: IsoFLOP curves from three published scaling law studies. Left: Chinchilla (training loss vs parameters). Center: Llama 3 (validation loss vs training tokens). Right: DeepSeek (bits-per-byte vs FLOPs/token). Each panel shows curves at multiple compute budgets, fit using Approach 2.

[3] https://docs.google.com/document/d/1Ab3cYJJYu7t_99Idj9gpHv2icZmRTPBJ_brJxsgnVA4

Published scaling law studies frequently report asymmetric loss surfaces. Some degree of asymmetry is nearly universal across modalities, and several reported exponent estimates reach the same levels of imbalance as the most extreme configuration in our simulations. The related biases documented in this paper therefore apply broadly.

Beyond surface asymmetry, the IsoFLOP curves in Figure 6 also show visible signs of off-center sampling and drift:

- **Off-center sampling:** At some compute budgets, the sampling grid does not appear centered at the curve minimum, placing more points on one side of the optimum than the other.
- **Drifting centers:** The degree of off-centering appears to vary across compute budgets rather than remaining constant, which is the drifting-bias pattern that distorts both exponents and intercepts.

To be clear, this is not a criticism of these studies. These are among the most careful and influential scaling law analyses yet published. The point is a more general one: the conditions under which Approach 2's biases take effect, asymmetric surfaces and imperfect sampling centers, appear to be the norm rather than the exception. The ideal conditions of Section 3 (symmetric surface, perfectly centered grids) occur rarely, if ever, in practice.

6.1 Compounding Errors

Given evidence that both surface asymmetry and off-center sampling are present in real studies, we can simulate what happens when these biases act simultaneously. Using the same three loss surfaces from earlier sections, we combine them with the 3× drift and 3× constant offset from the off-center analysis. We fit Approach 2 on compute budgets from 10^{17} to 10^{21} FLOPs and extrapolate D^* predictions to 10^{24} FLOPs across all four grid widths.

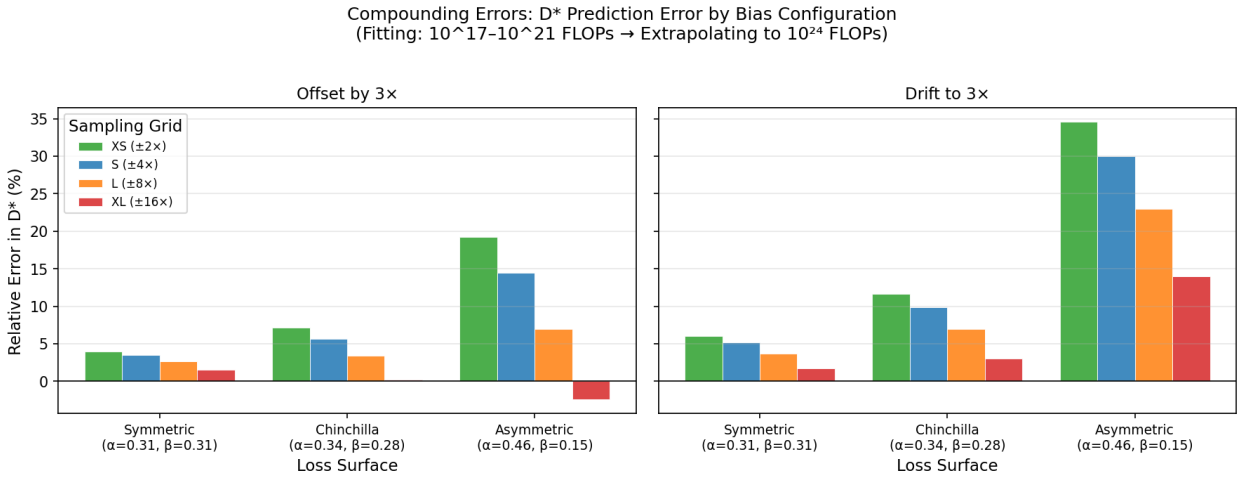


Figure 7: Relative error in D^* at 10^{24} FLOPs with off-center sampling on all three loss surfaces. Left: constant 3× center offset at every budget. Right: linear drift to 3× at the highest compute budget. Bars are grouped by sampling grid width (XS through XL). Negative values indicate underestimation; positive values indicate overestimation. On the symmetric surface, the constant offset results correspond to Figure 4 and the drift results correspond to Figure 5; the asymmetric surfaces reveal how these sampling biases interact with the inherent asymmetry bias. Raw data in Appendix B.

Comparing with the baseline in Figure 3, where asymmetry bias alone produces errors up to −5% on Chinchilla and −23% on the Asymmetric surface, the two bias sources interact in opposite

directions. Off-center sampling pushes errors positive (overestimating D^*), while asymmetry bias pushes errors negative (underestimating). When the two sources reinforce rather than cancel, errors grow substantially—reaching +35% at a maximum on the Asymmetric surface.

We believe these 3× perturbations are representative of realistic conditions. The IsoFLOP curves they produce (top-left panels of Figures 4 and 5) are qualitatively similar to those from the published studies in Figure 6, where sampling grids are not perfectly centered and the degree of off-centering varies across compute budgets.

Figure 12 in the appendix provides a more detailed view of these results. It shows how D^* extrapolation errors change across compute budgets, which sources of bias grow with extrapolation distance (drift), which remain constant (surface asymmetry and constant offsets), and how these patterns vary across two drift rates and center offset magnitudes.

Summary. Multiple bias sources act simultaneously in any real experiment. Surface asymmetry and off-center sampling each produce meaningful errors on their own. When they happen to act in the same direction, the combined error exceeds either one alone. On our hypothetical Asymmetric surface with drift to 3×, errors reach 35% even when using the narrowest grid, where the parabolic approximation is most accurate. When they oppose, partial cancellation can occur, but this depends on the specific combination of surface geometry, offset magnitude, and grid width.

7 Robust Fits: Unbiased Estimation with Linear Separation

The previous sections showed that Approach 2's parabolic approximation introduces systematic biases in intercepts (from asymmetry) and potentially exponents (from off-center sampling), and that the conditions driving these biases are visible in published scaling law studies. The natural alternative is Approach 3, which fits all five surface parameters (E, A, B, α, β) simultaneously via nonlinear least squares. This avoids the parabolic approximation but introduces other problems.

7.1 Problems with Direct Surface Fitting

A recent survey of over 50 scaling law papers [Moeini et al., 2025] documents the landscape of fitting practices and their failure modes. The problems described below apply to scaling law fitting in general, not just Chinchilla forms, but they are directly relevant because Approach 3 involves the same kind of nonlinear optimization. Over half of the papers surveyed do not fully specify their fitting procedure (optimizer, loss function, or initialization), which compounds reproducibility challenges.

The most common optimizers for scaling law fits are BFGS and L-BFGS. Some studies use SGD-family optimizers like Adam and Adagrad, though these are noted as sometimes poorly suited for curve fitting due to limited data efficiency. This is because methods like BFGS incorporate second-order derivatives (Hessian matrices) directly for greater data efficiency at the expense of memory utilization and scalability. Pure grid search offers an even more extreme tradeoff in this direction, which the survey notes as being employed by at least one study [Goyal et al., 2024].

A limitation common to all optimizers is sensitivity to initialization, which is primarily a consequence of "the difficulty of optimizing over this space, and the presence of many local minima." Common mitigations for this include grid search over hundreds or thousands of starting points (running the optimizer from each and keeping the best fit), random sampling of starting points, evaluating a coarse grid without optimization and seeding the optimizer from the single best candidate, or initializing from previously published parameter values.

Another significant issue is numerical stability, and this issue can be framed more concretely within the context of some of the simulations discussed in prior sections. For example, consider the Hessian of the residual sum of squares (RSS) objective for a five-parameter fit on noise-free data from the hypothetical Asymmetric surface ($\alpha = 0.465$, $\beta = 0.155$), using five IsoFLOP contours from 10^{17} to 10^{21} FLOPs with 15 points per curve. Its eigenvalues quantify how sensitive the objective is to perturbations along each parameter direction [Ghorbani et al., 2019], and the condition number κ (the ratio of the largest to the smallest eigenvalue) measures how difficult the landscape is for gradient-based methods to navigate. For this surface, the five eigenvalues span from approximately 8×10^{-6} to 3×10^{6}, giving $\kappa \approx 3.5 \times 10^{11}$. The two flattest directions (smallest eigenvalues) point almost entirely along the linear coefficients A and B. Near the optimum, perturbing either coefficient barely changes the RSS, making them effectively underdetermined by the data even when the data are perfect. The steepest directions are dominated by the scaling exponents α and β, and this difference in scale between exponents and linear coefficients is likely the primary source of numerical instability for most optimizers.

Quasi-Newton methods like L-BFGS build an approximate inverse Hessian to scale gradient steps across parameter directions. When eigenvalues span 12 orders of magnitude, the gradient signal along the flat A/B directions is negligible compared to the steep α/β directions, and convergence criteria are often satisfied by progress in the steep directions before the flat directions are resolved. Separating the linear parameters from the nonlinear search eliminates the ill-conditioned directions entirely. The resulting two-dimensional landscape over (α, β) has a Hessian condition number of $\kappa \approx 11$ in our example, a reduction by a factor of roughly 3×10^{10}. This motivates an algorithm that exploits the partially linear structure of the Chinchilla loss surface to search only the well-conditioned two-dimensional subspace.

7.2 Variable Projection (VPNLS)

The Chinchilla loss surface has a partially linear structure that can be exploited. For any fixed values of α and β, the remaining parameters (E, A, B) enter the model linearly and can be solved exactly via least squares. This is the same computational shortcut that motivates Approach 2 (optimizing exponential terms separately from linear terms), but applied here without the parabolic approximation.

The algorithm searches over (α, β) and, at each candidate pair, solves for (E, A, B) via least squares. A coarse 32×32 grid search identifies a good starting region, and a local optimizer refines it. The linear separation is maintained throughout. The optimizer only ever navigates the two-dimensional (α, β) surface, never the full five-parameter space. We term this method Variable Projection with Non-negative Least Squares (VPNLS).

In its simplest form, VPNLS uses non-negative least squares (NNLS) for the inner solve and a gradient-free optimizer like Nelder-Mead for the outer search:

```
function VPNLS(data):
  function objective(α, β):
    X ← [1, N^{-α}, D^{-β}]                          // design matrix
    (E, A, B) ← NNLS(X, L)                           // linear solve, E, A, B ≥ 0
    return ‖L − X · [E, A, B]‖²
  (α_0, β_0) ← arg min objective(α, β)               // coarse 32 × 32 grid
  (α*, β*) ← NelderMead(objective, start=(α_0, β_0)) // refine in 2D
  (E*, A*, B*) ← NNLS(X(α*, β*), L)                  // recover linear params
  return (E*, A*, B*, α*, β*)
```

Switching the inner solve from NNLS to ordinary least squares (OLS) makes the objective differentiable and enables analytical gradients. By the envelope theorem, the gradient of RSS with respect to (α, β) depends only on the current residuals and design matrix, not on implicit derivatives of the optimal (E, A, B). This allows L-BFGS-B to be used with exact gradients rather than finite differences:

```
function VPNLS(data):
  function objective_and_grad(α, β):
    X ← [1, N^{-α}, D^{-β}]                                  // design matrix
    (E, A, B) ← OLS(X, L)                                    // linear solve
    r ← L − X · [E, A, B]                                    // residual vector
    ∂RSS/∂α ← 2A · r^⊤ (log(N) ⊙ N^{-α})
    ∂RSS/∂β ← 2B · r^⊤ (log(D) ⊙ D^{-β})
    return ‖r‖², [∂RSS/∂α, ∂RSS/∂β]
  (α_0, β_0) ← arg min objective(α, β)                       // coarse 32 × 32 grid
  (α*, β*) ← L-BFGS-B(objective_and_grad, start=(α_0, β_0))  // α, β > 0
  (E*, A*, B*) ← OLS(X(α*, β*), L)                           // recover linear params
  return (E*, A*, B*, α*, β*)
```

Both variants achieve machine-precision parameter recovery on noise-free data. The gradient-free form with NNLS is simpler and easier to adapt to modified loss surface specifications. The analytical gradient form converges faster but requires deriving or computing gradients of the design matrix, which could also be handled by an automatic differentiation framework. The remainder of this study uses the gradient-based implementation as the primary VPNLS method.

Because the nonlinear search is over exponential terms only, grid density scales with the number of exponential terms rather than the total number of parameters. A 32^2 grid over (α, β) provides 1,024 candidates with fine resolution, whereas the same budget in 5D gives only $4^5 = 1{,}024$ points spread thinly across all parameters.

7.3 Method Comparison (Parameter Recovery)

We compare six method configurations on noise-free synthetic data across three loss surfaces (symmetric, Chinchilla, and high imbalance) and 20 sampling ranges.

The configurations fall into two groups. The first uses 5D direct optimization (Approach 3), fitting all five parameters jointly with L-BFGS-B using either analytical or numerical (3-point central difference) gradients. The second uses 2D variable projection over (α, β) only, comparing L-BFGS-B with analytical gradients, L-BFGS-B with numerical gradients, Nelder-Mead (gradient-free, NNLS inner solve), and a fine 256^2 grid search with no local refinement. Both groups use the same total initialization budget: the 5D methods search a $4^5 = 1{,}024$-point grid over all five parameters, while the 2D methods search a $32^2 = 1{,}024$-point grid over (α, β) only, so that accuracy differences reflect the optimizer and loss-landscape geometry rather than an initialization advantage.

All three locally optimized 2D variable projection methods (L-BFGS-B with analytical gradients, L-BFGS-B with numerical gradients, and Nelder-Mead) recover parameters to machine precision ($\sim 10^{-7}\%$ or better) across all surfaces. The well-conditioned 2D landscape makes even finite-difference gradients reliable in the reduced search space. High-resolution grid search (256^2) is stable but limited by grid resolution, providing the poorest precision among the 2D methods.

5D direct optimization (Approach 3) with analytical gradients also achieves high precision in

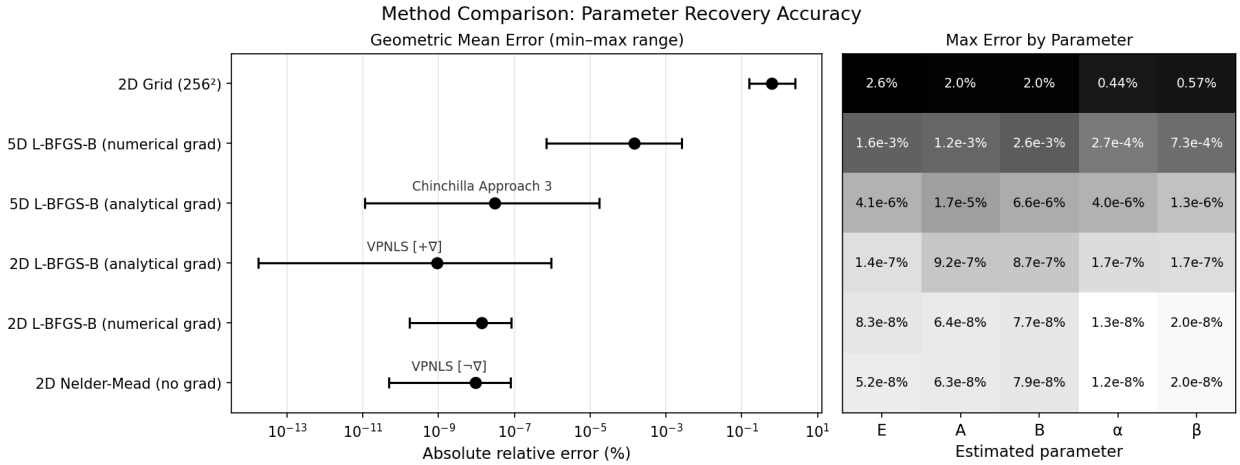


Figure 8: Comparison of six fitting methods on noise-free synthetic data across three loss surfaces and 20 sampling ranges (60 fits total per method). Left: geometric mean of |relative error| (%) pooled across all surfaces, grid widths, and parameters, with horizontal bars spanning the min-to-max range. Right: maximum |relative error| (%) per parameter, on a log-scale colormap. Methods are sorted by geometric mean error, with the worst at top. VPNLS [+∇] uses analytical gradients (L-BFGS-B with OLS inner solve); VPNLS [¬∇] is gradient-free (Nelder-Mead with NNLS inner solve).

this noiseless setting (errors on the order of $10^{-5}\%$), though with larger errors than any 2D method, particularly on the Chinchilla surface. The 5D numerical gradient variant performs worse, with errors on the order of $10^{-3}\%$. Figure 11 in the appendix breaks this down by surface and sampling range.

The full method comparison data is provided in Appendix C.

Summary. Variable projection reduces the nonlinear search to the exponential terms only, producing a well-conditioned 2D landscape where all tested optimizers converge reliably. The same initialization budget that spreads thinly across 5D ($4^5 = 1{,}024$ points) provides dense coverage in 2D ($32^2 = 1{,}024$ points). The exponents (α, β) are also confined to tight ranges (typically 0 to 1), unlike the linear coefficients which span orders of magnitude, making grid search over the reduced space especially effective.

7.4 Method Comparison (Exponent Inference)

The parameter recovery results above are noise-free, which is obviously not representative of practice. We now extend the compounding errors scenario (Section 6.1; Asymmetric surface with a 3× drift at the ±8× grid) to a statistical setting. Gaussian noise is added to loss values at three levels ($\sigma = 0.05, 0.1, 0.2$), the number of compute budgets varies from 2 to 4, and the number of points per IsoFLOP curve ranges from 4 to 32. Each configuration is repeated with 256 independent noise realizations, yielding 9,216 fits per method and 46,080 fits in total. Figure 13 in the appendix shows what the noisy IsoFLOP samples look like at each noise level.

Because a scaling law study is typically run once rather than repeated hundreds of times, sporadic optimizer failures that produce large errors in a minority of fits are arguably the most consequential practical risk. The comparison below emphasizes maximum errors alongside typical accuracy for this reason.

Five methods are compared. Approach 2 uses the parabolic IsoFLOP pipeline. Naive Ap-

proach 3 uses a single random initialization without LSE reparameterization. MLE Approach 3 uses grid initialization with the direct MSE objective (the maximum likelihood estimator under additive Gaussian noise). The final Approach 3 variant uses grid initialization with the LSE reparameterization and log-loss objective, matching the formulation in the Chinchilla paper [Hoffmann et al., 2022]. VPNLS uses L-BFGS-B with analytical gradients.

The parameter recovery comparison (Section 7.3) evaluated all five surface parameters, but Approach 2 does not estimate them individually. Here we focus on the scaling exponents $a = \beta/(\alpha+\beta)$ and $b = \alpha/(\alpha+\beta)$, which determine compute-optimal allocation and are the quantities all five methods can be compared on directly. Figure 9 pools these exponent errors across all experimental conditions.

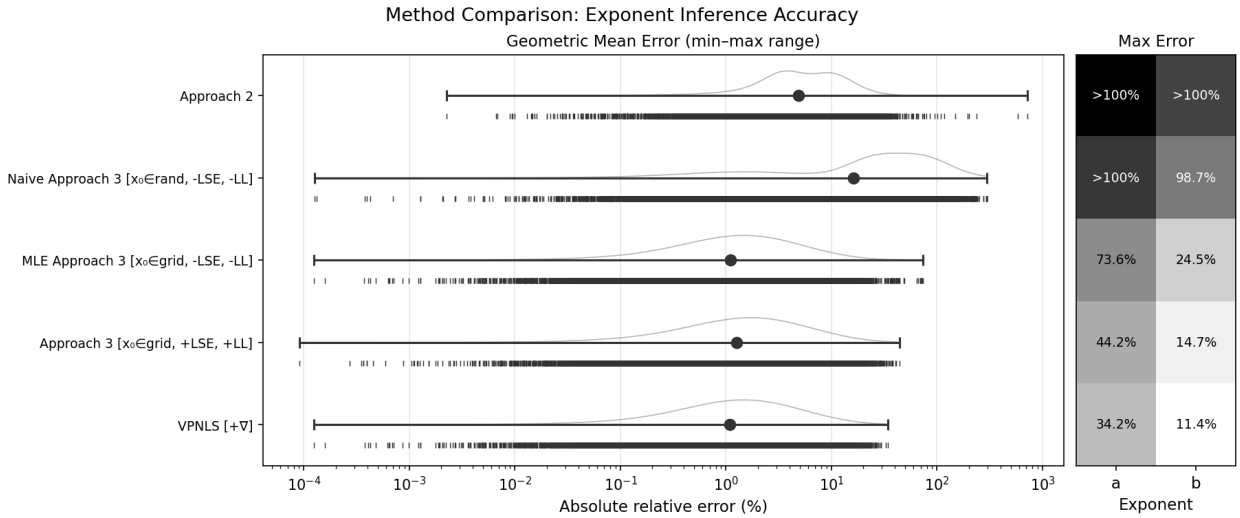


Figure 9: Comparison of five fitting methods on noisy synthetic data, pooled across three noise levels, three budget counts, four points-per-curve settings, and 256 noise realizations (9,216 fits per method). Left: geometric mean of |relative error| (%) with min-to-max range bars, rug ticks for individual errors, and kernel density estimate. Right: maximum |relative error| (%) per exponent. Methods sorted by worst-case error, worst at top. Surface: Asymmetric ($\alpha = 0.465$, $\beta = 0.155$) with 3× drift, ±8× grid, compute budgets 10^{17}–10^{21} FLOPs.

Method	GMean err%	Max a err%	Max b err%
VPNLS [+∇]	1.09	34.2	11.4
MLE Approach 3 [grid, −LSE, −LL]	1.11	73.6	24.5
Approach 3 [grid, +LSE, +LL]	1.26	44.2	14.7
Approach 2	4.84	715.5	238.5
Naive Approach 3 [rand, −LSE, −LL]	16.2	296.0	98.7

Table 5: Geometric mean and maximum |relative error| (%) on scaling exponents a and b, pooled across 3 noise levels × 3 budget counts × 4 points-per-curve settings × 256 seeds = 9,216 fits per method.

Approach 2 has the largest maximum errors overall (716% on a, 239% on b), reflecting the structural bias from the parabolic approximation (Section 4). Even with 32 points per curve and low noise, the systematic inaccuracy persists. Naive Approach 3 (single random initialization, no LSE) also produces extreme maximum errors (296% on a), confirming that without careful initialization and parameterization, 5D optimization is unreliable.

The canonical Chinchilla Approach 3 formulation (LSE reparameterization with log-loss objective, grid initialization) reduces maximum errors dramatically to 44% on a. The LSE reparameterization enforces positivity and stabilizes optimization. However, because the noise model in these simulations is additive Gaussian, the log-loss objective is slightly misspecified, which is visible as a higher geometric mean error compared to the MLE variant and VPNLS.

The MLE variant of Approach 3 (grid initialization, original-space MSE) is the correct maximum likelihood estimator for additive Gaussian noise. In this experiment, however, it exhibited larger maximum errors than either canonical Approach 3 or VPNLS (74% vs 44% vs 34% on the a exponent), despite slightly better typical accuracy. This is primarily due to the lack of LSE reparameterization.

VPNLS produced the smallest maximum errors in this experiment (34% on a, 11% on b) with typical accuracy roughly equivalent to Approach 3. The detailed breakdown in Figure 14 shows how errors distribute across noise levels, budget counts, and dataset sizes.

Summary. On asymmetric loss surfaces with realistic noise, Approach 2's structural bias persists regardless of data quality. Naive Approach 3 without careful initialization and parameterization performs even worse. With grid initialization and LSE reparameterization (as specified in the Chinchilla paper), Approach 3 achieves typical accuracy comparable to VPNLS. VPNLS produced the smallest maximum errors in this experiment, though all well-configured direct fitting methods substantially outperform Approach 2.

8 Error Costs: Misallocation at Frontier Scale

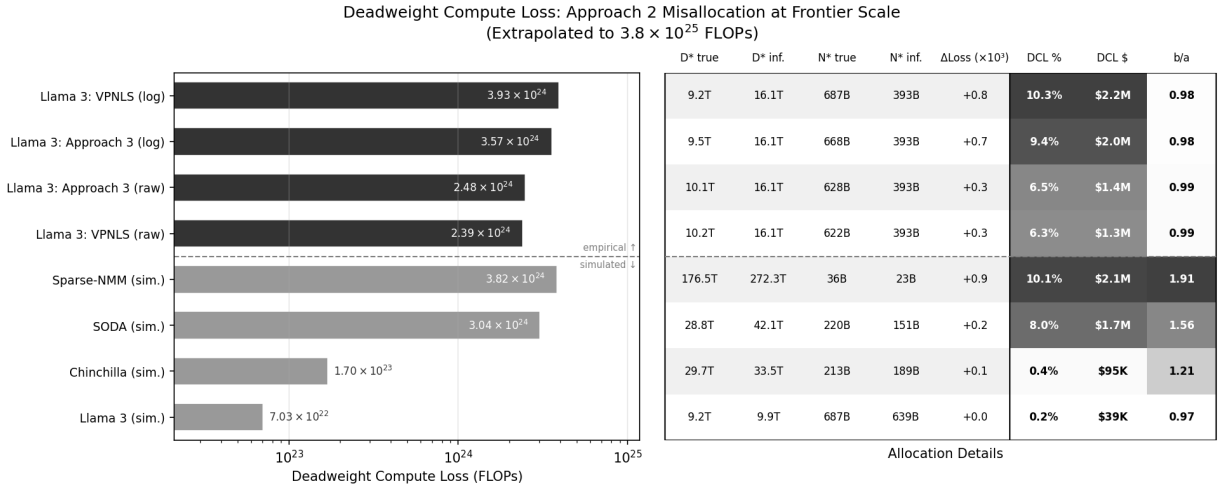
Scaling laws serve at least two broad purposes in practice. At smaller scales they guide research decisions like comparing architectures, datasets, and training strategies before committing to a full run. For production models of small to medium size, allocation planning typically targets deliberate overtraining (training on more data than compute-optimal) rather than compute-optimality itself, and Approach 3 or variations on it are much more common in that setting. The compute-optimal scaling laws studied in this paper are most directly relevant to the largest models in frontier families, which are still frequently trained near compute-optimality. For these runs, the biases documented in the preceding sections may translate directly into wasted compute.

To quantify this, we assess the cost of Approach 2 misallocations at the compute scale of Llama 3 405B (3.8×10^{25} FLOPs) [Grattafiori et al., 2024], which remains one of the most compute-intensive open models trained as of early 2026 [Epoch AI, 2026]. We also examine three additional loss surfaces with published Approach 3 fit statistics—the original Chinchilla surface [Hoffmann et al., 2022], and two multimodal models for which scaling is substantially more asymmetric, SODA [Manakul et al., 2026] (interleaved audio-text) and Sparse-NMM [Shukor et al., 2025b] (native multimodal with sparse mixture-of-experts). These surfaces span a range of exponent asymmetry ratios (β/α) from near-unity to nearly 2:1, providing a view of how the biases compound as the loss surface departs from the symmetric case.

We measure misallocation in terms of **Deadweight Compute Loss** (DCL), defined as the excess FLOPs consumed by training at the Approach 2–inferred allocation relative to the budget that would achieve the same loss under optimal allocation. Given a compute budget C and an inferred data allocation D^*, the model size is forced by the constraint $N = C/(6D^*)$. The resulting loss $L(N, D^*)$ is suboptimal. Inverting the optimal loss function $L_{\text{opt}}(C)$ yields the smaller budget C_{eq} that would reach this same loss with optimal allocation; $\text{DCL} = C - C_{\text{eq}}$. This is analogous to deadweight loss in economics, where dataset and model sizes define a supply–demand tradeoff

that maximizes a "total surplus"—or equivalently, minimizes validation loss—only at the compute-optimal allocation. Any departure from optimality wastes compute in the same sense that a price distortion wastes economic surplus.

For Llama 3, we digitize individual IsoFLOP data points from Figure 2 of the original paper [Grattafiori et al., 2024] following the SVG-coordinate extraction method of Besiroglu et al. [Besiroglu et al., 2024], then fit both Approach 2 and two Approach 3 implementations (VPNLS and direct L-BFGS-B) to obtain competing allocations at the target budget. For Chinchilla, SODA, and Sparse-NMM we take a simulated approach instead, generating synthetic IsoFLOP data from each published loss surface, fitting Approach 2 to it under the same bias assumptions used in earlier sections (XS grid width, 3× sampling drift), and comparing the resulting extrapolations. We include a simulated Llama 3 condition as well, which provides an implicit measure of how much of the empirical Llama 3 error is attributable to the real IsoFLOP experiment design versus the biases we introduce systematically. Dollar costs are computed assuming 50% model FLOPs utilization [Sardana et al., 2024] and $2 per H100-hour [Ettinger et al., 2025].



D* true	D* inf.	N* true	N* inf.	ΔLoss (×10³)	DCL %	DCL $	b/a
9.2T	16.1T	687B	393B	+0.8	10.3%	$2.2M	0.98
9.5T	16.1T	668B	393B	+0.7	9.4%	$2.0M	0.98
10.1T	16.1T	628B	393B	+0.3	6.5%	$1.4M	0.99
10.2T	16.1T	622B	393B	+0.3	6.3%	$1.3M	0.99
176.5T	272.3T	36B	23B	+0.9	10.1%	$2.1M	1.91
28.8T	42.1T	220B	151B	+0.2	8.0%	$1.7M	1.56
29.7T	33.5T	213B	189B	+0.1	0.4%	$95K	1.21
9.2T	9.9T	687B	639B	+0.0	0.2%	$39K	0.97

Figure 10: Approach 2 misallocation costs extrapolated to 3.8×10^{25} FLOPs. Left: Deadweight Compute Loss (DCL) as a percentage of budget. Right: allocation details including true vs. inferred token counts and model sizes, loss penalty, and dollar cost. "Empirical" rows use Approach 2 power laws fit to digitized Llama 3 IsoFLOP data, evaluated against VPNLS and Approach 3 surfaces. "Simulated" rows use synthetic IsoFLOP data generated from each published surface with XS grid and 3× drift bias.

Figure 10 shows the results. On the empirical Llama 3 data, Approach 2 overestimates the compute-optimal token count by 58–75% depending on the reference surface, yielding DCL of 6–10% of budget ($1.3–2.2M). The simulated Llama 3 condition, which isolates the systematic biases from the IsoFLOP experiment design, produces only 0.2% DCL ($39K) at $\beta/\alpha \approx 0.97$. This indicates that most of the empirical error is likely attributable to the particular shape of the real IsoFLOP data rather than to the biases modeled here. Chinchilla ($\beta/\alpha \approx 1.21$) allocation errors remain similarly small at 0.4% ($95K). The multimodal surfaces show substantially higher costs as exponent asymmetry increases: SODA ($\beta/\alpha \approx 1.56$) reaches 8.0% DCL ($1.7M) and Sparse-NMM ($\beta/\alpha \approx 1.91$) reaches 10.1% ($2.1M). For domains with highly asymmetric scaling exponents, which are more common in multimodal and non-language settings, Approach 2 misallocations can waste a non-trivial fraction of a large training budget.

Summary. Based on Llama 3 IsoFLOP data, Approach 2 overestimates the compute-optimal token count by 58–75% at 3.8×10^{25} FLOPs, wasting 6–10% of the training budget ($1.3–2.2M in H100 time). Multimodal loss surfaces, where exponent asymmetry is typically larger, exhibit similarly significant misallocation errors (8–10% of budget costing $1.7–2.1M) under the same simulation conditions for which Llama 3 errors reach only 0.2% waste. This suggests that designing IsoFLOP experiments to avoid Approach 2 biases is particularly important in those settings, or that Approach 2 should not be used to fit data from them.

9 Conclusion

The Approach 2 biases documented in this paper are structural, not statistical. They exist on noise-free data with perfect experimental conditions and, as the noisy method comparison (Section 7.4) shows, persist under realistic noise levels with varying amounts of data.

At least three sources of Approach 2 error can compound in practice. IsoFLOP sampling grid width, uncentered IsoFLOP sampling, and loss surface asymmetry all bias inference and extrapolations in different ways. Published IsoFLOP curves from prominent scaling law studies (Section 6) show clear signs of both asymmetry and off-center sampling. At frontier compute scales (Section 8), we estimate that these biases translate to a potential 6–10% decrease in training FLOPs ($1.3–2.2M) on Llama 3 data and potentially more on multimodal surfaces with greater asymmetry.

Approach 3 eliminates these biases. With grid initialization and LSE reparameterization (as specified in the original Chinchilla paper [Hoffmann et al., 2022]), it achieves typical accuracy comparable to VPNLS in our experiments. A recent survey [Moeini et al., 2025] suggests that these important optimization details may be omitted or not reported in some studies.

VPNLS is at least as stable and accurate as more typical Approach 3 implementations while being structurally simpler. Variable projection separates exponential from linear terms, reducing the nonlinear search to the exponential terms only. This makes dense grid search practical because the grid grows with the number of exponential terms rather than all parameters, and these exponents occupy tight ranges (typically 0 to 1) unlike the linear coefficients which span several orders of magnitude.

This structural advantage extends naturally to richer loss surface specifications. The analytical extensions discussed in the introduction (Section 1), such as epochs, data quality, and MoE sparsity, often add linear terms that could be omitted from direct optimization. This structural simplification may provide a more scalable foundation for future work. It also lends well to simple implementations of the Chinchilla loss model alone, e.g. it is possible to express a simplified version of the algorithm[4] that guarantees globally optimal solutions at a desired level of scaling exponent precision, uses only ∼70 lines of JavaScript code, and requires no dependencies (no optimization or linear algebra libraries).

Lastly, practitioners using Approach 2 should be aware that scaling exponent estimates carry a systematic bias that often grows with loss surface asymmetry, sampling center offsets, and sampling grid width. When fitting surfaces directly with Approach 3 instead, grid initialization and LSE reparameterization should be used. And VPNLS offers equivalent accuracy with a simpler optimization structure to aid in scaling this approach to richer formulations, or as a simpler alternative to Approach 2.

[4] https://gist.github.com/eric-czech/ebd9a80d58c7b5e9c40ba390ff884617

9.1 Limitations

Several limitations scope the conclusions of this study.

- **Irreducible loss dominance at large scale.** At sufficiently large compute budgets, scaling properties are dominated entirely by the irreducible loss E. When token counts and model sizes at fixed compute budgets are large enough, the Chinchilla surface reaches E asymptotically and all training configurations become equally effective, meaning that extrapolations are irrelevant and compute-optimal training is no longer informed by scaling laws. We assume this study is only relevant to practitioners working in a regime where downstream model quality can still effectively be informed by scaling law extrapolations per the Chinchilla model.
- **Assumed correctness of the Chinchilla loss surface.** We assume the Chinchilla loss surface model $L(N, D) = E + A/N^{\alpha} + B/D^{\beta}$ is correct in practice. While there is substantial evidence legitimizing this model [Schaeffer et al., 2025], alternatives exist, including the Kaplan loss model [Kaplan et al., 2020], refined analytical surfaces like Farseer [Li et al., 2025] and MuPT [Yang et al., 2024], and agent-discovered functional forms [Lin et al., 2025].
- **Qualitative characterization of published study errors.** Likely errors in published studies are characterized qualitatively rather than quantified. We believe the qualitative characterization is compelling enough on its own to justify that real IsoFLOP sampling pathologies occur in practice, but they are difficult to quantify precisely because they do not follow the convenient theoretical model we use for those pathologies in our simulations.

References

Ibrahim M. Alabdulmohsin, Behnam Neyshabur, and Xiaohua Zhai. Getting ViT in shape: Scaling laws for compute-optimal model design. In *NeurIPS*, 2023.

Lucas Bandarkar et al. ATLAS: Adaptive transfer scaling laws for multilingual pretraining, fine-tuning, and decoding the curse of multilinguality. *arXiv preprint arXiv:2510.22037*, 2025.

Tamay Besiroglu, Ege Erdil, Matthew Barnett, and Josh You. Chinchilla scaling: A replication attempt. *arXiv preprint arXiv:2404.10102*, 2024.

Akshita Bhagia, Jiacheng Liu, Alexander Wettig, David Heineman, Oyvind Tafjord, Ananya Harsh Jha, Luca Soldaini, Noah A. Smith, Dirk Groeneveld, Pang Wei Koh, Jesse Dodge, and Hannaneh Hajishirzi. Establishing task scaling laws via compute-efficient model ladders. *arXiv preprint arXiv:2412.04403*, 2024.

Xiao Bi, Deli Chen, Guanting Chen, Shanhuang Chen, Damai Dai, Chengqi Deng, et al. DeepSeek LLM: Scaling open-source language models with longtermism. *arXiv preprint arXiv:2401.02954*, 2024.

Yuda Bi and Vince D. Calhoun. Scaling laws are redundancy laws. *arXiv preprint arXiv:2509.20721*, 2025.

Yilun Chen et al. Scaling laws of motion forecasting and planning – technical report. *arXiv preprint arXiv:2506.08228*, 2025.

Chen Cheng and Aurelien Lucchi. Theory of scaling laws for in-context regression: Depth, width, context and time. *arXiv preprint arXiv:2510.01098*, 2025.

Epoch AI. Open models compute threshold. https://epoch.ai/data-insights/open-models-threshold, 2026.

Allyson Ettinger, Amanda Bertsch, Bailey Kuehl, et al. OLMo 3: Open language models at scale. *arXiv preprint arXiv:2512.13961*, 2025.

Behrooz Ghorbani, Shankar Krishnan, and Ying Xiao. An investigation into neural net optimization via hessian eigenvalue density. *arXiv preprint arXiv:1901.10159*, 2019.

Sachin Goyal et al. Scaling laws for data filtering – data curation cannot be compute agnostic. In *CVPR*, 2024.

Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, et al. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.

Siddhant Haldar and Lerrel Pinto. Scaling laws for imitation learning in single-agent games. *Transactions on Machine Learning Research*, 2023.

Tatsunori Hashimoto et al. Representation matters: Assessing the importance of subgroup allocations in training data. *arXiv preprint arXiv:2103.03399*, 2021.

Samyak He, Valentin Hofmann, Hailey Schoelkopf, Sumanth Dathathri, Shruthi Bannur, and Ethan Dyer. Joint MoE scaling laws: Mixture of experts can be memory efficient. *arXiv preprint arXiv:2502.05172*, 2025.

Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, et al. Training compute-optimal large language models. *arXiv preprint arXiv:2203.15556*, 2022.

Ethan Jiang et al. Exploring scaling laws for EHR foundation models. *arXiv preprint arXiv:2505.22964*, 2025.

Tian Jin, Ahmed Imtiaz Humayun, Utku Evci, Suvinay Subramanian, Amir Yazdanbakhsh, Dan Alistarh, and Gintare Karolina Dziugaite. The journey matters: Average parameter count over pre-training unifies sparse and dense scaling laws. *arXiv preprint arXiv:2501.12486*, 2025.

Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*, 2020.

Tanishq Kumar et al. Scaling laws for precision. *arXiv preprint arXiv:2411.04330*, 2024.

Houyi Li, Wenzhen Zheng, Qiufeng Wang, Zhenyu Ding, Haoying Wang, Zili Wang, Shijie Xuyang, Ning Ding, Shuigeng Zhou, Xiangyu Zhang, and Daxin Jiang. Predictable scale: Part II, Farseer: A refined scaling law in large language models. *arXiv preprint arXiv:2506.10972*, 2025.

Zhengyang Li et al. Scaling laws for diffusion transformers. *arXiv preprint arXiv:2410.08184*, 2024.

Haowei Lin, Haotian Ye, Wenzheng Feng, Quzhe Huang, Yujun Li, Hubert Lim, Zhengrui Li, Xiangyu Wang, Jianzhu Ma, Yitao Liang, and James Zou. Can language models discover scaling laws? *arXiv preprint arXiv:2507.21184*, 2025.

Potsawee Manakul, Woody Haosheng Gan, Martijn Bartelds, Guangzhi Sun, William Held, and Diyi Yang. Scaling open discrete audio foundation models with interleaved semantic, acoustic, and text tokens. *arXiv preprint arXiv:2602.16687*, 2026.

Armin Moeini et al. (mis)fitting: A survey of scaling laws. In *ICLR*, 2025.

Niklas Muennighoff, Alexander M. Rush, Boaz Barak, Teven Le Scao, Aleksandra Piktus, Nouamane Tazi, Sampo Pyysalo, and Thomas Wolf. Scaling data-constrained language models. *arXiv preprint arXiv:2305.16264*, 2023.

Eric Nguyen, Michael Poli, Matthew G. Durrant, et al. Sequence modeling and design from molecular to genome scale with Evo. *bioRxiv preprint 2024.02.27.582234*, 2024.

Shen Nie et al. Scaling behavior of discrete diffusion language models. *arXiv preprint arXiv:2512.10858*, 2025.

Nikhil Sardana and Jonathan Frankle. Reconciling Kaplan and Chinchilla scaling laws. *Transactions on Machine Learning Research*, 2024.

Nikhil Sardana, Jacob Portes, Sasha Doubov, and Jonathan Frankle. Beyond chinchilla-optimal – accounting for inference in language model scaling laws. *arXiv preprint arXiv:2401.00448*, 2024.

Rylan Schaeffer, Noam Levi, Andreas Kirsch, Theo Guenais, Brando Miranda, Elyas Obbad, and Sanmi Koyejo. Evaluating the robustness of chinchilla compute-optimal scaling. *arXiv preprint arXiv:2509.23963*, 2025.

Mustafa Shukor, Louis Bethune, Dan Busbridge, David Grangier, Enrico Fini, Alaaeldin El-Nouby, and Pierre Ablin. Scaling laws for optimal data mixtures. *arXiv preprint arXiv:2507.09404*, 2025a.

Mustafa Shukor, Enrico Fini, Victor Guilherme Turrisi da Costa, Matthieu Cord, Joshua Susskind, and Alaaeldin El-Nouby. Scaling laws for native multimodal models. *arXiv preprint arXiv:2504.07951*, 2025b.

Anirudh Subramanyam, Yuxin Chen, and Robert L. Grossman. Scaling laws revisited: Modeling the role of data quality in language model pretraining. *arXiv preprint arXiv:2510.03313*, 2025.

Yi Tay et al. Training compute-optimal transformer encoder models. In *EMNLP*, 2025.

Changxin Tian, Kunlong Chen, Jia Liu, Ziqi Liu, Zhiqiang Zhang, and Jun Zhou. Towards greater leverage: Scaling laws for efficient mixture-of-experts language models. *arXiv preprint arXiv:2507.17702*, 2025.

Alexandra Volkova, Mher Safaryan, Christoph H. Lampert, and Dan Alistarh. Towards robust scaling laws for optimizers. *arXiv preprint arXiv:2602.07712*, 2026.

Neeraj Wagh et al. Scaling laws for compute optimal biosignal transformers. 2024.

Xinghua Yang et al. MuPT: A generative symbolic music pretrained transformer. *arXiv preprint arXiv:2404.06393*, 2024.

A Extrapolation Error Raw Data

Surface	Grid	True D^*	Inferred D^*	Abs Error	Rel Error
Symmetric ($\alpha = \beta = 0.31$)					
	XS ($\pm 2\times$)	408.2B	408.2B	≈ 0	$\approx 0\%$
	S ($\pm 4\times$)	408.2B	408.2B	≈ 0	$\approx 0\%$
	L ($\pm 8\times$)	408.2B	408.2B	≈ 0	$\approx 0\%$
	XL ($\pm 16\times$)	408.2B	408.2B	≈ 0	$\approx 0\%$
Chinchilla ($\alpha = 0.34$, $\beta = 0.28$)					
	XS ($\pm 2\times$)	4.04T	4.02T	-13.2B	-0.33%
	S ($\pm 4\times$)	4.04T	3.98T	-52.5B	-1.30%
	L ($\pm 8\times$)	4.04T	3.92T	-117.2B	-2.90%
	XL ($\pm 16\times$)	4.04T	3.83T	-205.8B	-5.10%
Asymmetric ($\alpha = 0.465$, $\beta = 0.155$)					
	XS ($\pm 2\times$)	45.1Q	44.3Q	-755T	-1.67%
	S ($\pm 4\times$)	45.1Q	42.2Q	-2.9Q	-6.50%
	L ($\pm 8\times$)	45.1Q	38.8Q	-6.3Q	-13.91%
	XL ($\pm 16\times$)	45.1Q	34.7Q	-10.4Q	-23.12%

Table 6: Extrapolation error raw data underlying Figure 3. Training range: 10^{17}–10^{21} FLOPs; evaluation budget: 10^{24} FLOPs. B = billion, T = trillion, Q = quadrillion.

B Compounding Errors Raw Data

Config	Surface	Grid	True D^*	Inferred D^*	Rel Error
Offset 3× (sampling center at 3× true optimum at every budget)					
	Symmetric	XS	408.2B	424.5B	+3.97%
	Symmetric	S	408.2B	422.4B	+3.47%
	Symmetric	L	408.2B	419.0B	+2.65%
	Symmetric	XL	408.2B	414.4B	+1.51%
	Chinchilla	XS	4.04T	4.32T	+7.11%
	Chinchilla	S	4.04T	4.27T	+5.69%
	Chinchilla	L	4.04T	4.17T	+3.38%
	Chinchilla	XL	4.04T	4.05T	+0.24%
	Asymmetric	XS	45.1Q	53.8Q	+19.22%
	Asymmetric	S	45.1Q	51.6Q	+14.41%
	Asymmetric	L	45.1Q	48.2Q	+6.96%
	Asymmetric	XL	45.1Q	44.0Q	−2.42%
Drift to 3× (center drifts from true optimum to 3× at highest budget)					
	Symmetric	XS	408.2B	433.0B	+6.07%
	Symmetric	S	408.2B	429.4B	+5.17%
	Symmetric	L	408.2B	423.3B	+3.70%
	Symmetric	XL	408.2B	415.1B	+1.69%
	Chinchilla	XS	4.04T	4.50T	+11.61%
	Chinchilla	S	4.04T	4.43T	+9.83%
	Chinchilla	L	4.04T	4.32T	+6.94%
	Chinchilla	XL	4.04T	4.16T	+3.05%
	Asymmetric	XS	45.1Q	60.7Q	+34.57%
	Asymmetric	S	45.1Q	58.7Q	+30.04%
	Asymmetric	L	45.1Q	55.5Q	+22.97%
	Asymmetric	XL	45.1Q	51.4Q	+14.00%

Table 7: Compounding errors raw data underlying Figure 7. Training range: 10^{17}–10^{21} FLOPs; evaluation budget: 10^{24} FLOPs. B = billion, T = trillion, Q = quadrillion.

C Method Comparison Data

Method	Failures	Max E	Max A	Max B	Max α	Max β
2D L-BFGS-B (analytical)	0/60	1.4×10^{-7}	9.2×10^{-7}	8.7×10^{-7}	1.7×10^{-7}	1.7×10^{-7}
2D L-BFGS-B (numerical)	0/60	8.3×10^{-8}	6.4×10^{-8}	7.7×10^{-8}	1.3×10^{-8}	2.0×10^{-8}
2D Nelder-Mead (VPNLS)	0/60	5.2×10^{-8}	6.3×10^{-8}	7.9×10^{-8}	1.2×10^{-8}	2.0×10^{-8}
2D Grid (256^2)	0/60	2.58	2.03	2.03	0.44	0.57
5D L-BFGS-B (analytical)	0/60	4.1×10^{-6}	1.7×10^{-5}	6.6×10^{-6}	4.0×10^{-6}	1.3×10^{-6}
5D L-BFGS-B (numerical)	0/60	1.6×10^{-3}	1.2×10^{-3}	2.6×10^{-3}	2.7×10^{-4}	7.3×10^{-4}

Table 8: Method comparison: maximum |relative error| (%) per parameter across 60 fits (3 surfaces × 20 sampling ranges). All methods converged on all fits. Methods sorted by precision (best to worst).

D Detailed Method Comparison

Full per-parameter, per-surface, per-sampling-range error breakdown for all six method configurations.

Figure 11: Detailed method comparison: absolute relative error vs sampling range for all six configurations. Rows correspond to loss surfaces (symmetric, Chinchilla, Asymmetric); columns correspond to parameters (E, A, B, α, β).

E Combined Extrapolation Error by Compute Budget

Detailed view of D^* extrapolation error as a function of compute budget, showing how errors evolve from 10^{22} to 10^{25} FLOPs across sampling ranges, loss surfaces, and bias configurations.

F IsoFLOP Samples with Noise

G Exponent Inference Error Breakdown

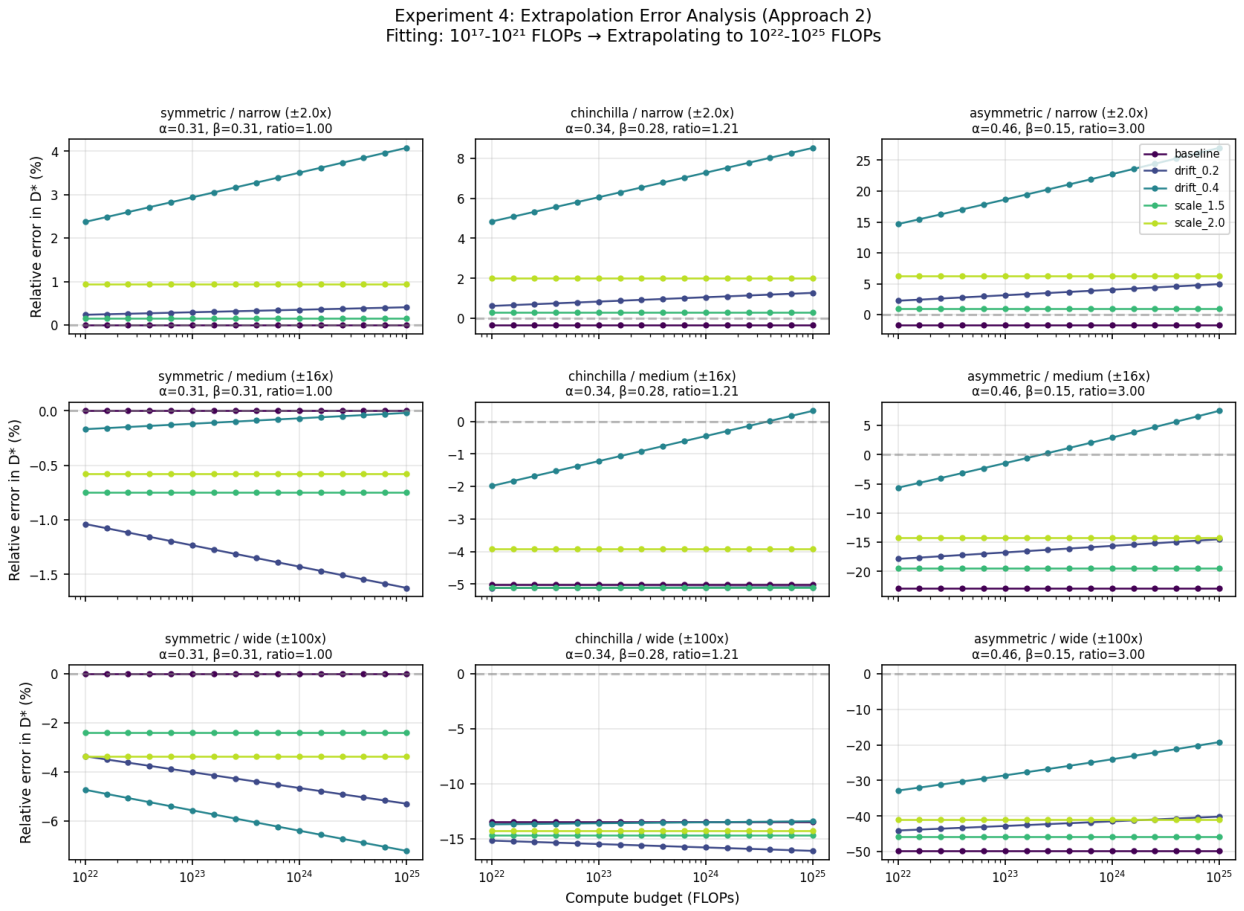


Figure 12: Combined extrapolation error by compute budget. Rows correspond to sampling ranges (narrow, medium, wide); columns correspond to loss surfaces (symmetric, Chinchilla, Asymmetric). Each panel shows relative D^* error vs extrapolation compute budget with one curve per bias configuration.

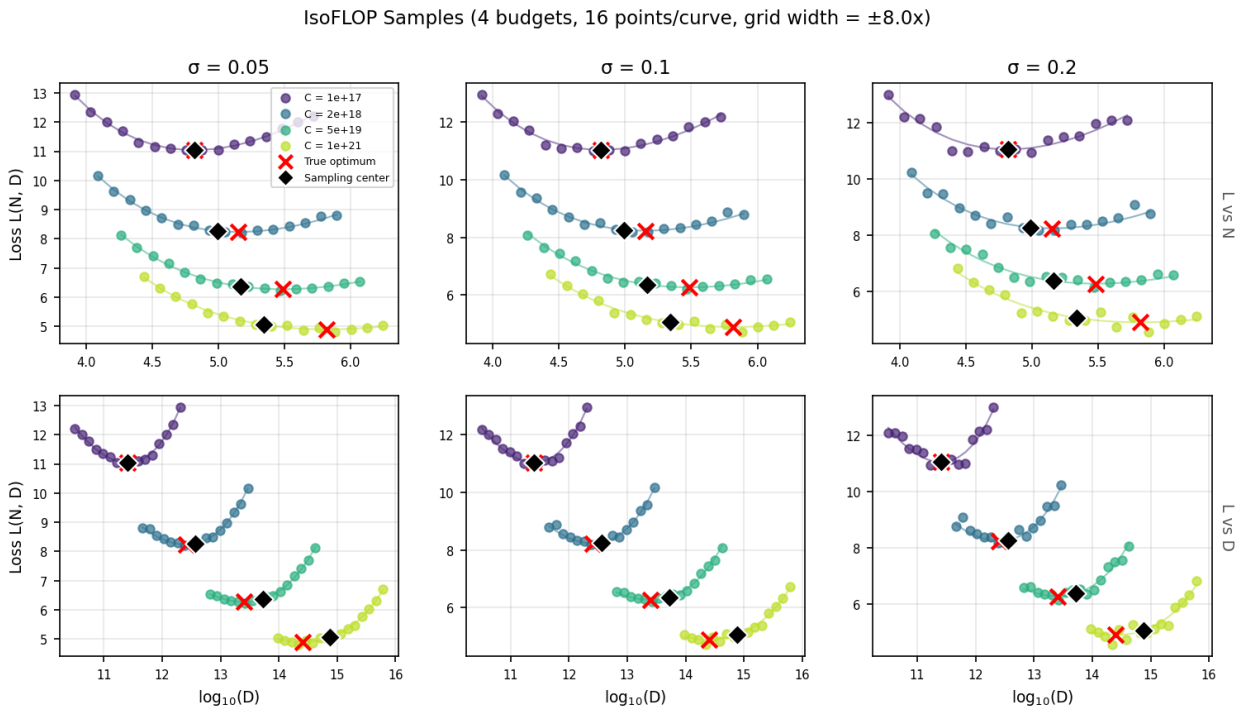


Figure 13: Noisy IsoFLOP samples used in the exponent inference comparison (Figure 9). Columns correspond to noise levels ($\sigma = 0.05, 0.1, 0.2$); rows show loss versus $\log_{10}(N)$ (top) and $\log_{10}(D)$ (bottom). Scatter points are noisy observations; solid curves show the noiseless reference surface. Red $\times$ marks the true compute-optimal point at each budget; black diamonds mark the sampling centers, which drift away from the true optima at higher compute budgets. Surface: Asymmetric ($\alpha = 0.465$, $\beta = 0.155$), 4 budgets (10^{17}–10^{21} FLOPs), 32 points per curve, $\pm 8\times$ grid.

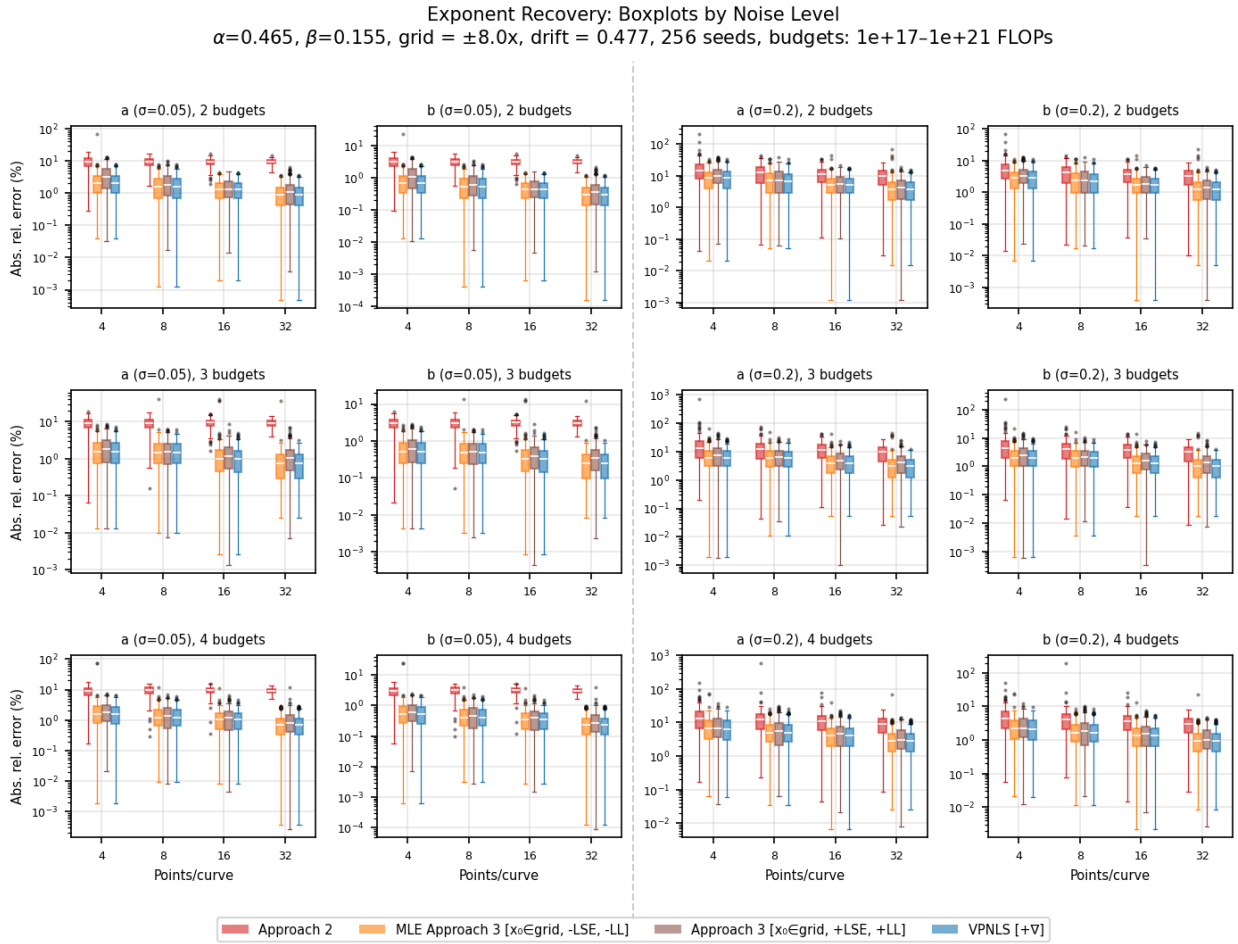


Figure 14: Per-condition breakdown of exponent inference errors from Figure 9, showing 4 of 5 methods (excludes Naive Approach 3, which serves only as a negative control in the main figure). Rows correspond to number of compute budgets (2, 3, 4); left columns show the lowest noise level ($\sigma = 0.05$) and right columns show the highest ($\sigma = 0.2$), each split by exponent (a and b). Within each panel, boxplots show absolute relative error (%) on a log scale for each method at each points-per-curve setting (4, 8, 16, 32), over 256 noise realizations.